

DSA Tutorial - 4

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Topic: Data base using struct

INPUT :

```
#include <stdio.h>
#include <string.h>
```

```
struct Student {
    char name[50];
    int rollno;
    float cgpa;
};
```

```
struct Student db[100];
int count = 0;
```

```
void create();
void display();
void append();
void search();
void modify();
void sort();
```

```
int main() {
    int choice;
```

```
    while(1) {
        printf("\n=== Student Database
Menu ===\n");
        printf("1. Create Database\n");
        printf("2. Display Database\n");
        printf("3. Append (Add 1
Student)\n");
```

```
        printf("4. Search (By Roll
No)\n");
        printf("5. Modify (By Roll
No)\n");
        printf("6. Sort (By Roll No)\n");
        printf("7. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
```

```
        switch(choice) {
            case 1: create(); break;
            case 2: display(); break;
            case 3: append(); break;
            case 4: search(); break;
            case 5: modify(); break;
            case 6: sort(); break;
            case 7: printf("Exiting
program...\n"); return 0;
            default: printf("Invalid choice!
Try again.\n");
```

```
        }
    }
    return 0;
}
```

```
void create() {
    printf("How many students do you
want to add? ");
    scanf("%d", &count);
    for(int i = 0; i < count; i++) {
```

```

        printf("\nEnter details for student
%d:\n", i + 1);
        printf("Name: ");
        scanf("%s", db[i].name);
        printf("Roll No: ");
        scanf("%d", &db[i].rollno);
        printf("CGPA: ");
        scanf("%f", &db[i].cgpa);
    }
    printf("Database created
successfully!\n");
}

void display() {
    if (count == 0) {
        printf("Database is empty!\n");
        return;
    }
    printf("\n%-15s %-10s %-10s\n",
"NAME", "ROLL NO", "CGPA");

    printf("-----\n");
    for(int i = 0; i < count; i++) {
        printf("%-15s %-10d %-10.2f\n",
db[i].name, db[i].rollno, db[i].cgpa);
    }
}

void append() {
    if (count >= 100) {
        printf("Database is full!\n");
        return;
    }
    printf("\nEnter details for new
student:\n");
    printf("Name: ");

```

```

        scanf("%s", db[count].name);
        printf("Roll No: ");
        scanf("%d", &db[count].rollno);
        printf("CGPA: ");
        scanf("%f", &db[count].cgpa);
        count++;
        printf("Student added
successfully!\n");
    }

void search() {
    int key, found = 0;
    printf("Enter Roll No to search: ");
    scanf("%d", &key);

    for(int i = 0; i < count; i++) {
        if(db[i].rollno == key) {
            printf("\nRecord Found!\n");
            printf("Name: %s\nRoll No:
%d\nCGPA: %.2f\n", db[i].name,
db[i].rollno, db[i].cgpa);
            found = 1;
            break;
        }
    }
    if(!found) {
        printf("Student with Roll No %d
not found.\n", key);
    }
}

void modify() {
    int key, found = 0;
    printf("Enter Roll No of the student
to modify: ");
    scanf("%d", &key);

```

```

        for(int i = 0; i < count; i++) {
            if(db[i].rollno == key) {
                printf("\nCurrent details ->
Name: %s, CGPA: %.2f\n",
db[i].name, db[i].cgpa);
                printf("Enter new Name: ");
                scanf("%s", db[i].name);
                printf("Enter new CGPA: ");
                scanf("%f", &db[i].cgpa);
                printf("Record updated
successfully!\n");
                found = 1;
                break;
            }
        }
        if(!found) {
            printf("Student with Roll No %d
not found.\n", key);
        }
    }

void sort() {
    struct Student temp;
    for(int i = 0; i < count - 1; i++) {
        for(int j = 0; j < count - i - 1; j++)
        {
            if(db[j].rollno > db[j+1].rollno) {
                temp = db[j];
                db[j] = db[j+1];
                db[j+1] = temp;
            }
        }
    }
    printf("Database sorted
successfully by Roll No!\n");
}

```

OUTPUT :

```

=== Student Database Menu ===

```

1. Create Database
2. Display Database
3. Append (Add 1 Student)
4. Search (By Roll No)
5. Modify (By Roll No)
6. Sort (By Roll No)
7. Exit

```

Enter your choice: 2

```

NAME	ROLL NO	CGPA

harsh	12	9.00